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Soil and water conservation in India: Strategies and research challenges

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ABSTRACT

Soil and water are the most important natural resources that meet all human needs, and protect the environment and the civilization. Ever increasing demands for land and water resources due to burgeoning population cause the degradation of these two precious natural resources. Land degradation is a serious problem in India and out of 329 million hectare (Mha) of total geographical area, 146.8 Mha is reported as degraded. India is losing a huge amount of money in terms of loss of productivity from these degraded lands. Though India receives good amount of annual rainfall (1190 mm), due to enhanced need for domestic, agriculture and industrial developmental activities, over-exploitation and laxity of water resource, it becomes scarce. The declining trend of groundwater level in all parts of the country also indicates that the assured supply of good quality water will become a concern for country's development. Hence holistic management approach of soil and water resources by linking social and economic development with protection of natural ecosystem is the need of the hour. Creation of inventory on soil and water resources, use of modern technological tools for developing conservation strategies, changes in policy to protect natural resources, development of site specific soil and water conservation measures and involvement of local people in conserving the soil and water resources are the future challenges for the scientists and policy makers. This paper highlights the soil and water conservation strategies and areas to be addressed on priority.

Key words: Climate change, Groundwater, Land degradation, Soil and water conservation, Water resources

INTRODUCTION

Soil and water are the most important natural resources and physical base for all life supporting system as water sustains life where soil acts as a medium. These two resources are too precious for the mankind as they meet all the needs and protect the environment as well as the civilization. Land degradation has become a serious problem in both rainfed as well as irrigated areas of India. India is losing a huge amount of money from these degraded lands. Cost lost due to land degradations were documented by various authors (Joshi and Agnihotri, 1984; Parikh and Ghosh, 1995; Joshi *et al.*, 1996; Srinivasarao, 2013) in terms of declining crop productivity, land use intensity, changing cropping patterns, high input use and declining profit. Loss of production in India at Rupees 68 billion in 1988-1989 was estimated by Reddy (2003) using the National Remote Sensing Agency dataset. Additional losses resulting from salinization, alkalinisation and water logging were estimated as Rs 8 billion. Of late, in a comprehensive study made on the impact of water erosion on crop productivity, it was revealed that soil erosion due to water resulted in an annual crop production loss of 13.4 Mt in cereal, oil seeds and pulse crops (Sharda *et al.*, 2010).

Out of a total reported geographical area of 329 Mha of India, about 146.8 Mha are degraded by various factors. Water erosion and wind erosion together account for 70% of the total degraded land and the remaining 30% is due to salinity, acidity and combination of other factors. Annual soil loss of India is estimated as 5334 Mt along with these 8.4 Mt of major nutrients also lost (Prasad and Biswas, 2000). Annually, 2052 million tonnes of soil is carried by rivers out of this nearly 480 million tonnes is being deposited in various reservoirs which results in 1–2% loss of storage capacity per year (Dhruva Narayana and Rambabu, 1983). According to National Commission on Agriculture, reservoirs in India are silting up at a rate of three to four times faster than the designed rates. It is a matter of concern that out of 329 Mha geographical area, about 145 Mha is under cultivation and there is no scope to bring more area under cultivation.

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Singh (1999) reported that the cropping intensity at present is only 136%. The per capita cultivable land will reduce to 0.14 ha in 2025 AD compared to 0.33 ha in 1950.

The burgeoning population and their impact in every sphere of development, *viz*. agriculture, industry and urbanization depend primarily on water resources leading to ever increasing demands of water. Increasing water demand, over exploitation of ground water resources and inefficiency of tapping the surface water and also harvesting rain water created an imbalance resulting into dearth of water availability for sustainable food production and the domestic usage. FAO has reported that the global water withdrawal increased from less than 600 km^3 $year^{-1}$ in 1900 to almost 4000 km^3 $year^{-1}$ in 2010. Further, it is assessed that it will increase to 5100 km^3 during 2025 with a rise of 8.4-12.2% from the current rate of withdrawal. In 1995, 76% of the world population had water availability of less than 5100 m^3 per annum per capita and it is predicted that in 2025, most of the Earth's population will be living under low water supply. It is projected that the situation of water user categories in the world will not shift much excepting increase in water consumption due to urbanization in the developing countries. The water consumption for agriculture will be around 70%, industry around 20% and residential and commercial around 10%. Due to the effects of climate change and uncertain rainfall the use of water in agriculture will increase by the expansion of irrigated land. By late 1970, almost all developed and developing countries started intensive irrigation development to ensure increased crop production. Subsequently, the global rate of increase in irrigated areas has slowed down primarily due to very high cost of construction of irrigation system and soil degradation problems. Thus lots of efforts are required to harness the water resource potential and to combat the drought and flood in the country. The management of soil and water resources needs holistic approach by linking socio economic developmental activities with eco-friendly environment.

LAND DEGRADATION

An estimate says that 430 Mha land is subjected to severe land degradation in the world (Barrow 1991) and the impact of degradation in Asia will be around 18%. In India, high percentage of area under degraded land in Jammu and Kashmir (69.24%), Himachal Pradesh (50.90%) is due to snow cover and degraded forest; Nagaland (22.37%), Manipur (59.01%), Mizoram (21.20%) is due to shifting cultivation, Sikkim (53.67%) is due to degraded forest and in Rajasthan (29.64%) is due to sandy area. The category-wise distribution of wastelands shows that highest percentage (4.76%) belongs to the category 'land with scrub' mainly distributed in the southern states of India followed by 'underutilized forestland' (3.42%) distributed throughout the country. Among the 583 districts, 19 districts mainly distributed in NE states, Jammu Kashmir and Rajasthan have got more that 50% of geographical area under wastelands (MoRD & NRSA, 2005).

The severity and extent of soil degradation in the country has been previously assessed by many agencies (Table 1). According to the National Bureau of Soil Survey and Land Use Planning, 146.8 Mha land is degraded in India. Water and wind erosion are the most serious degradation problem in India, resulting in loss of topsoil and terrain deformation followed by acidity and water logging (Fig 1). Based on first approximation analysis of existing soil loss data, the average soil erosion rate

Table 1. Extent of soil degradation in India as assessed by different organizations

S.No	Organizations	Degraded area (M ha)
1.	National Commission on Agriculture (NCA, 1976)	148.1
2.	Ministry of Agriculture-Soil and Water Conservation Division (MOA, 1978)	175.0
3.	Department of Environment (Vohra, 1980)	95.0
4.	National Watershed Development Board (NWDB, 1985)	123.0
5.	Society for Promotion of Wastelands Development (Bhumbla and Khare, 1984)	129.0
6.	National Remote Sensing Agency (NRSA, 1985)	53.3
7.	Ministry of Agriculture (MOA, 1985)	173.6
8.	Ministry of Agriculture (MOA, 1994)	107.4
9.	National Bureau of Soil Survey and land Use Planning (NBSS&LUP, 1994)	187.7
10.	National Bureau of Soil Survey and land Use Planning, Revised (NBSS&LUP, 2005)	146.8

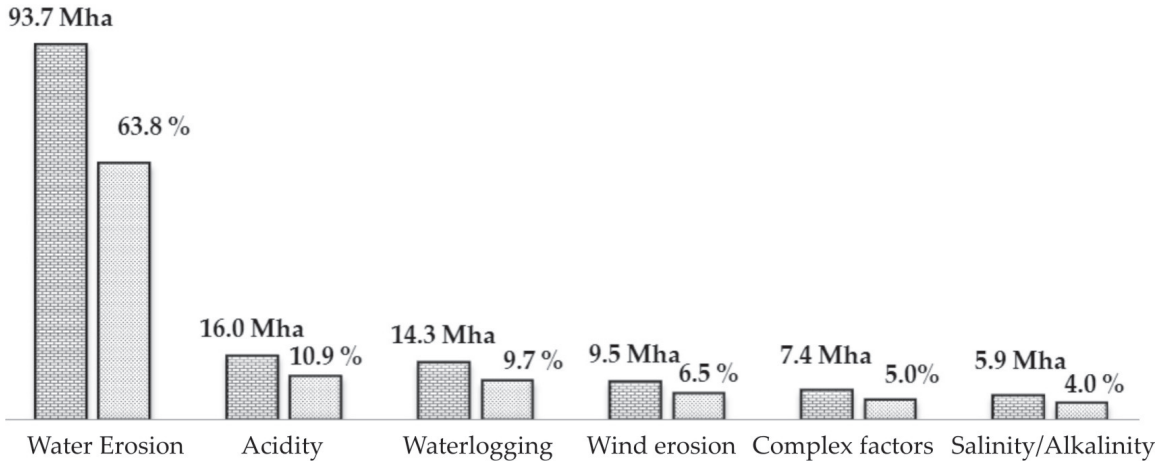


Fig. 1. Contribution of land degradation factors in India
(*Source*: NBSS&LUP, 2005)

was 16.4 ton ha^{-1} $year^{-1}$, resulting in an annual total soil loss of 5.3 billion tons throughout the country. Nearly 29% of total eroded soil is permanently lost to the sea, while 61% is simply transferred from one place to another and the remaining 10% is deposited in reservoirs.

ISSUES ON WATER RESOURCES

India has a large and intricate network of river systems of which the most prominent are the Himalayan river systems draining the major plains of the country. Apart from this, numerous water bodies present in the sub-continent make it one of the wettest places in the world after South America. The annual precipitation including snowfall, which is the main source of the water in the country, is estimated to be 4000 billion cubic metres (BCM). Central Water Commission estimated the basin wise resource potential of the country, which occurs as natural runoff in the rivers is about 1869 cu.km. Ganga-Brahmaputra-Meghna system is the major contributor to total water resources potential of the country with about 60% of the share in total water resources potential of the various rivers. In majority of river basins, present utilization is significantly high and is in the range of 50–95% of utilizable surface resources except the rivers such as Narmada and Mahanadi where the utilization is quite low.

Groundwater resources consists of two parts, namely, dynamic and static, the dynamic resource present in the zone of water table fluctuation, whereas the static resource will be below the dynamic zone which remains perennially saturated. As per the National Water Policy- 2002, the dynamic groundwater resource is essentially the exploitable quantity of groundwater, which is recharged annually, and is also termed as replenishable groundwater resource. The annual replenishable groundwater resource of the country is 431 BCM and the net groundwater availability is 396 BCM after allocating 35 BCM for natural discharges during non-monsoon season. The annual groundwater draft for the year 2009 was 243 BCM, out of which 221 BCM is utilized for irrigation and 22 BCM is used for domestic and industrial purposes. The overall stage groundwater development of the country is estimated as 61% (CGWB, 2014). However, 1615 assessment units fall under semi-critical (550), critical (226) and over exploited (839) category, indicating that the groundwater resources in these areas is already being developed, more than that is being recharged. Further areas of the 30 assessment units are completely covered by saline groundwater. The contribution of rainfall to annual replenishable groundwater resource of the country is 67% (Chatterjee and Purohit, 2009), indicating the dependence on rainfall for recharge of ground water resources.

Groundwater recharge is significantly high is the Indo-Gangetic-Brahmaputra alluvial belt, where the rainfall is plenty and thick piles of unconsolidated alluvial formations favours recharge. Recharge per ha of area in these regions varies from 0.28 to 1.35 m. The coastal alluvial belt also has relatively high recharge, in the range of 0.16 to 0.40 m. whereas the annual recharge is only 0.10 m in Western India with arid climate. Similar is the case with major part of the southern peninsular India (Chatterjee and Purohit, 2009). The Ministry of water resources estimated the ultimate irrigation potential created in India till 2014 as 140 Mha. In that 76 Mha is from surface water resources and the remaining 64 Mha by ground water resources (CWC, 2014).

CLIMATE CHANGE IMPACT ON WATER RESOURCES

Some of the studies carried out in the Indian Himalayas clearly point out on an increase in glacial melt (Kumar *et al.*, 2007). For instance Baspa basin of Himachal Pradesh has shown an increase in the winter stream flow by 75% as compared to 1966. It is estimated that Himalayan Mountains cover surface area of permanent snow and ice in the region is about 97,020 km^2 with 12,930 km^2 volumes. In these mountains, it is estimated that 10 to 20% of the total surface area is covered by glaciers while an additional area ranging from 30 to 40% has seasonal snow cover. This snow and glacial-melt water keeps our Indian rivers perennial throughout the year. Bahadur (1999) reported that a very conservative estimate gives at least 500 km^3/yr from snow and ice melt water contributions to Himalayan streams, while about 515 km^3/yr from the upper mountains. The most useful facet of glacial runoff is the fact that glaciers release more water in a drought year and less water in a flood year and thus ensuring water supply even during the lean years. The snow line and glacier boundaries are sensitive to changes in climatic conditions. Almost 67% of the glaciers in the Himalayan mountain ranges have retreated in the past decade (Ageta and Kadota, 1992; Yamada *et al.*, 1996). The mean equilibrium line altitude at which snow accumulation is equal to snow ablation for glacier is estimated to be about 50-80 meters higher relative to the altitude during the first half of the 19th Century.

Available records suggest that Gangotri glacier is retreating about 28 meters per year. Increase in temperature due to global warming is likely to increase the melting far more rapidly than the accumulation. According to IPCC (1998) glacial melt is expected to increase under changed climate conditions, which would lead to increased summer flows in some river systems for few decades, followed by a reduction in flow as the glaciers disappear.

In India floods also one of the climate change extreme which affects 10 million hectares of area every year of which 50 % of the area is under agricultural land. Rashtriya Barh Ayog (RBA) constituted by the Government of India in 1976 carried out an extensive analysis to estimate the flood-affected area in the country and reported that 40 Mha areas is liable to floods (Mall *et al.*, 2006). According to CWC (2002), as many as 99 districts spread over 14 states were identified as drought prone districts in the country. Most of the drought prone areas are concentrated in the states of Rajasthan, Karnataka, Andhra Pradesh, Maharashtra and Gujarat. Human factors that influence drought include demand for water through population growth and agricultural practices, and modification of land use that directly influences the storage conditions and hydrological response of catchments and thus its vulnerability to drought (WMO, 2002).

SOIL AND WATER CONSERVATION STRATEGIES IN INDIA

In India, it is not only important to increase the availability of water but also efficient utilization of water should be promoted by adopting other techniques like multiple use of water, proper soil and crop management, low cost micro irrigation, integrated farming system, etc. Adaptation of suitable soil and water conservation measures and cultivating crops based on land suitability is the first and most important factor which can reduce the soil degradation problems. Micro scale rain water harvesting from the farm lands and multiple uses in the farmers field itself is the right approach for increasing the water productivity and livelihood security of rural farmers. The various degraded land categories have to be treated using the available technologies as below:

Gullied or ravenous land

The key to development of gullied and or ravenous land lies in the management of runoff. Prevention of sheet or rill erosion on the tablelands and in transition zones is the first step to control the further increment of gullied/ravenous lands. Preventing damage from river back flow, promoting afforestation and grass land development with involvement of local peoples and government departments and encourage the farmers to go for suitable horticultural plantation in the susceptible regions are the other feasible measures for rehabilitation of gullied / ravenous land. Wherever vegetative measures are not feasible or economical appropriate engineering / mechanical measures have to be adopted or both the measures can be combined suitably. Combined use of engineering measures like trench-cum-bund and vegetative barrier of *Vetveria zizanioides* and sambuta (*Saccharum Spp*) reduced runoff and soil losses to half of that observed in control plots and thereby increased test crop (finger millet) yield by 18–33% (Dass *et al.*, 2010). Also, the vegetative filter-

strips that are typically much wider (more than 5 m) can be established between field borders and water ways (Blanco-Canqul *et al.,* 2004) to protect land and reduce pollution in the water bodies (Srivastava *et al.*, 1998).

Land with or without scrub

Degraded rocky / stony and gravelly upland and shallow hardpan soil areas with very scanty and poor distribution of unpalatable shrubs and grasses as well as low production potential, are recommended for development of silvipastures. Degraded lands when converted to perennial land use like forestlands ensures minimum soil disturbance and enhance soil organic carbon sequestration, thereby reduces the risk of soil erosion (Lal, 2015). Improved soil and water conservation measures including vegetative barriers along with rainwater harvesting techniques are important pre-requisite to establish such production systems.

Water logging and salinization

Water logging is the main problem in the areas of soils with poor drainage capacity. Water logged areas can be efficiently managed by adopting efficient water application technology and modern irrigation system, raising intercrops with integrated nutrient management and adopting crop rotation. In addition vertical drain, sub-surface drain, bio-drain and lining of canal will helps to drain the excess water from these areas.

Degraded forest land

Partly degraded forest areas can be improved by gap filling, reseeding and transplanting with suitable plant species using soil and water conservation measures such as trenching and contour furrowing. These kinds of degraded forest areas should be properly protected through fencing so that seeds already existing will germinate and the condition of the forestland can be improved. The ground flora should be enriched by growing suitable shrub and grass species in contour furrows. Grazing in these areas should be restricted and properly regulated with the involvement of local community. Regular monitoring of developmental activities should be ensured and timely action should be taken to prevent degradation and development of good forest cover. Joint Forest Management is one of the excellent system which aids in the regeneration of degraded forests with the help of the nearby villagers who are dependent on the forest for their livelihood (Binodini, 2016). Co-operation from nearby villages should also be taken to ensure social protection of these forest areas.

Degraded pasture / grazing lands

Degraded grazing lands are mainly distributed in the states of Rajasthan, Haryana etc., and at present almost all grazing lands, especially village commons are in degraded condition. Loss of ground cover has a bearing on infiltration rate causing low moisture storage in soil profile. These lands need priority for rehabilitation and can be achieved with the co-operation and active participation of local people. Degraded vegetation can be regenerated by providing suitable soil and moisture conservation measures and the protection. Inder Dev *et al.* (2016) reported that introduction of fodder trees and grasses/legumes in grazing lands in combination with trenches reduced the runoff to 10.2% compared to 40.5% where there is no soil and water conservation measures and enhanced the biomass production potential of the pasture lands. The approach should be ecologically suitable and acceptable to local people.

Shifting cultivation

Shifting cultivation is a common agriculture practice in the hill regions of India and it is extensively practiced in North Eastern states, Orissa and the Eastern Ghats with an estimated forest area of about 4.35 Mha. Initially the fallow period followed in shifting cultivation is more than 30 years, but now the pressure on agricultural land has increased significantly due to burgeoning population and the fallow period has reduced to 2-3 years. The short fallow period between two cultivation is not sufficient for the natural processes of recuperation of the disturbed ecosystem leading to decline of soil fertility and land degradation. The land degraded by shifting cultivation has to be rehabilitated by adopting silvipasture, silviculture and agri-horticulture system. Lenka *et al.* (2012, 2013) reported that in eastern ghats of Odisha, shifting cultivation ravaged hills could be successfully rehabilitated with production-cum-conservation based agroforestry systems consisting of Guava / cashew + *Stylosanthes hamata* + Trench system.

Shifting and stabilized dunes

Shifting barchan dunes due to their inherent vulnerability become quite active during summer

season and sands starts moving and engulfing the adjoining agricultural lands and engineering structures. This kind of dunes needs immediate attention for stabilization and control of degradation. The stabilization measure includes providing protection from biota, micro wind breaks and re-vegetation in the affected areas. Due to intense biotic activities the top, crest and upper part of the flanks of most of the parabolic dunes are usually covered with loose sand. In such sites sand movement can be checked to a maximum extent by planting suitable grasses and plant species. *Lasiurus sindicus* locally known as 'sewan grass' is used as a fodder grass for the desert livestock of Thar desert also used for stabilizing the blowing sand dunes (Sinha, 1996). Singh and Rathod (2002) recommended sowing of *Prosopis juliflora* in sand dunes for early stabilization. *Cassia angustifolia* and *Cenchrus ciliaris* or the other local grasses of suitable combinations also can help to improve the biomass production and biodiversity in these kinds of sandy areas.

FUTURE THRUST AREAS

In India water resource potential is estimated to be 187 M ha m of which utilizable water resource potential is 110 M ha m comprising 60 M ha m and 50 M ha m surface and ground water resources. Indian region is highly sensitive to climate change and demand for water from groundwater may increase if precipitation decreases and surface water inflows decrease, this leads to a decrease in discharge elsewhere. Being a riverine country, 316 m ha is drained by major rivers and their tributaries. Major area of the country suffers from water induced soil erosion and thus acute in the country. Hence it is essential to arrest the menace of soil loss following the hierarchy of streams and rivers. Thus lots of efforts are required to control the land degradation and harness the water resource potential and to combat the drought and flood in the country. The identified researchable areas for future are prioritized and categorized as below:

Policy implications

State wise policy implications needs to be developed on surface and ground water exploitation for agriculture, domestic and industrial use to avoid over exploitation of water resources. Appropriate state wise strategies for augmentation of ground water recharge and its quantification and management needs to be derived for sustainable ground water availability. Deriving strategies for corporate social responsibilities for managing ground water resources will be another area where we can enhance the ground water recharge.

Inventories on soil degradation

Soil degradation status needs to be up dated periodically as the land degradation status is changing frequently. Inventory, data base and characterization with respect to erosion hazards needs to linked with remote sensing and GIS techniques enable to update them as and when required. Remote sensing and GIS based tools and procedures are to be developed for erosion assessment. Researchers need to concentrate on development of area specific soil loss tolerance limits and land quality indices for maintenance of sustainability. Effect of forest vegetation and biotic disturbances on natural resource conservation for sustainable production needs to be studied.

Soil conservation

In spite of various technologies available for conserving soil resources, still scope to develop the technologies for soil health management including rehabilitation soil degraded by chemical actions. Location specific technologies for improving crop and fruit productivity through moisture conservation, silvipastural and agroforestry practices needs to be developed. Refinement of bioengineering measures including application of natural fibre based Geotextiles needs to be undertaken for rehabilitation of mine environments and landslide affected slopes in various parts of the country.

Water management

Studies on monitoring water availability and resource conservation impacts of watershed management programmes needs to be strengthen across the country. Indigenous sensor based equipments needs to be developed for monitoring surface and ground water resources. Priority should be given to identify, evaluate and refinement the water harvesting techniques in different agro ecological regions of the country to mitigate the climate change impact on water resources. Studies on upstream downstream linkages at micro and macro regional scales for assessing impact of watershed management programmes on flow regimes (surface and ground water resources) and perenniality of streams and rivers needs to be initiated. Design procedures and nomographic solutions for stream bank protection structures

based on climatic and physiographic features is another area to be addressed on priority.

Climate change impact

Observational data and data access network to assess the recent experience in climate variability and extreme events, impacts of projected climate change and variability and associated hydrological events in India at river basin / aquifer level needs to be strengthened. Impact of climate change on groundwater aquifers, including quality, recharges rates, and flow dynamics should be quantified. Immediate research studies to be initiated on impact of sea level rise on the coastal groundwater regime, vulnerability of regional water resources to climate change and identifying key risks and prioritizing adaptation responses.

Participatory water resources management

Participatory water resource management is another area where user's participation is lacking in management of water resources. Hence, the constraints in managing water resources by users or beneficiaries needs to be assessed and appropriate procedure should be derived at national level for participatory water resources management.

CONCLUSIONS

Out of 329 Mha geographical area in India, about 45 % of the land (146.8 Mha) is estimated as degraded. Adopting appropriate mitigation strategies for rehabilitation of such degraded lands are the urgent need of the country for sustainable production. Water resource potential of India is to the extent of 187 M ham of which utilizable water resource potential is only 110 M ha m including surface and ground water resources. As our Indian region is highly sensitive to climate change the decrease in precipitation results in decreased surface water inflows and increase the demand for ground water. Derivation of appropriate policy implications for surface and ground water management, creation of remote sensing and GIS based inventories on land degradation, development of indigenous sensor based instruments for monitoring hydrological parameters, studies on climate change impact on soil and water resources, refinement of soil and moisture conservation and water harvesting structures to mitigate the climate change impacts and participatory water resources management conserving the soil and water resources are the future challenges for the scientists and policy makers.

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